

From the plate-resonator equations to the ERP

- 1 Plate with a point force and three sprung-mass resonators

$$D\nabla^4 w + \rho h \ddot{w} = f_d(t) \delta(x - x_d) \delta(y - y_d) - \sum_{i=1}^3 \delta(x - x_i) \delta(y - y_i) [k_i(w - u_i) + c_i(\dot{w} - \dot{u}_i)]$$

$$m_i \ddot{u}_i + c_i(\dot{u}_i - \dot{w}) + k_i(u_i - w) = 0, \quad w_i = w(x_i, y_i, t), \quad i = 1, 2, 3$$

$$D = \frac{Eh^3}{12(1 - \nu^2)}, \quad k_i = m_i (2\pi f_{t,i})^2$$

- 2 Modal expansion of the plate (simply supported edges)

$$w(x, y, t) = \sum_{m,n} q_{mn}(t) \phi_{mn}(x, y), \quad \phi_{mn} = \frac{2}{\sqrt{\rho h L_x L_y}} \sin \frac{m\pi x}{L_x} \sin \frac{n\pi y}{L_y}$$

$$\omega_{mn} = \sqrt{\frac{D}{\rho h} \left[\left(\frac{m\pi}{L_x} \right)^2 + \left(\frac{n\pi}{L_y} \right)^2 \right]}, \quad N \text{ modes } (15 \times 10, \text{ or } 6 \times 3 \text{ for the 18-mode data})$$

- 3 Coupled modal system (plate modes and the three resonators)

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{f}, \quad \mathbf{x} = [q_1, \dots, q_N, u_1, u_2, u_3]^\top, \quad \mathbf{M} = \text{diag}(1, \dots, 1, m_1, m_2, m_3)$$

$$\mathbf{K}_{qq} = \text{diag}(\omega_{mn}^2) + \sum_i k_i \vec{\phi}_i \vec{\phi}_i^\top, \quad \mathbf{K}_{qu_i} = -k_i \vec{\phi}_i, \quad \mathbf{K}_{u_i u_i} = k_i, \quad \vec{\phi}_i = [\phi_{mn}(x_i, y_i)]$$

$$\mathbf{C}: \text{the same with } c_i \text{ in place of } k_i \text{ and without the } \omega_{mn}^2 \text{ term; } \mathbf{f} = [F_0 \phi_{mn}(x_d, y_d), 0, 0, 0]^\top$$

- 4 Harmonic response ($e^{j\omega t}$, $\omega = 2\pi f$)

$$(\mathbf{K} + j\omega \mathbf{C} - \omega^2 \mathbf{M})\hat{\mathbf{x}}(\omega) = \mathbf{f} \Rightarrow \hat{q}_{mn}(\omega), \hat{u}_i(\omega)$$

- 5 Velocity, radiated power and ERP

$$\hat{w}(x, y, \omega) = \sum_{m,n} \hat{q}_{mn} \phi_{mn}, \quad \hat{v} = j\omega \hat{w}$$

$$P(f) = \frac{1}{2} \rho_L c_L \iint |\hat{v}|^2 dA = \frac{1}{2} \rho_L c_L \omega^2 \hat{\mathbf{q}}^H \mathbf{G} \hat{\mathbf{q}}, \quad G_{ij} = \iint \phi_i \phi_j dA$$

$$\text{ERP}(f) = 10 \log_{10} \frac{P(f)}{P_{\text{ref}}}$$